Qualcomm Technologies, Inc.

SW6100 Memory Guide

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1 Memory documentation

Review memory configurations, optimize software, customize and test DRAM components, troubleshoot issues, and monitor memory usage.

1.1 Memory overview

Dynamic random-access memory (DRAM) configuration

Understand the DRAM capabilities and memory configuration.

Enable DRAM

Enable DRAM software to start the memory subsystem.

1.2 Customize and validate the DRAM components

Customize DRAM

Find DRAM components in the Qualcomm preferred vendor list (PVL) component database.

1.3 Troubleshoot DRAM issues

DRAM initialization

Log entry and description of DRAM initialization during the XBL stage of bootup.

DRAM usage

Track memory usage in real-time during execution.

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2 Memory overview

Qualcomm chip products use low-power double data rate (LPDDR) dynamic random-access memory (DRAM) to support system operations. DRAM is a type of volatile memory that stores data frequently accessed by the processor and requires constant refreshing to retain information.

The Qualcomm memory controller supports JEDEC - compliant LPDDR5 and LPDDR5X DRAM components. You can select the type of DRAM based on your device requirements.

2.1 DRAM capabilities of SW6100

The memory controller supports high-speed LPDDR5/LPDDR5X SDRAM in single-channel package-on-package (PoP) configurations.

- LPDDR5/LPDDR5X SDRAM supports a maximum memory density of 4 GB.
- LPDDR5/LPDDR5X SDRAM operates at a clock speed of up to 4200 MHz clock speed (1 × 16-bit).

The following figure shows the DRAM memory channel architecture for SW6100:

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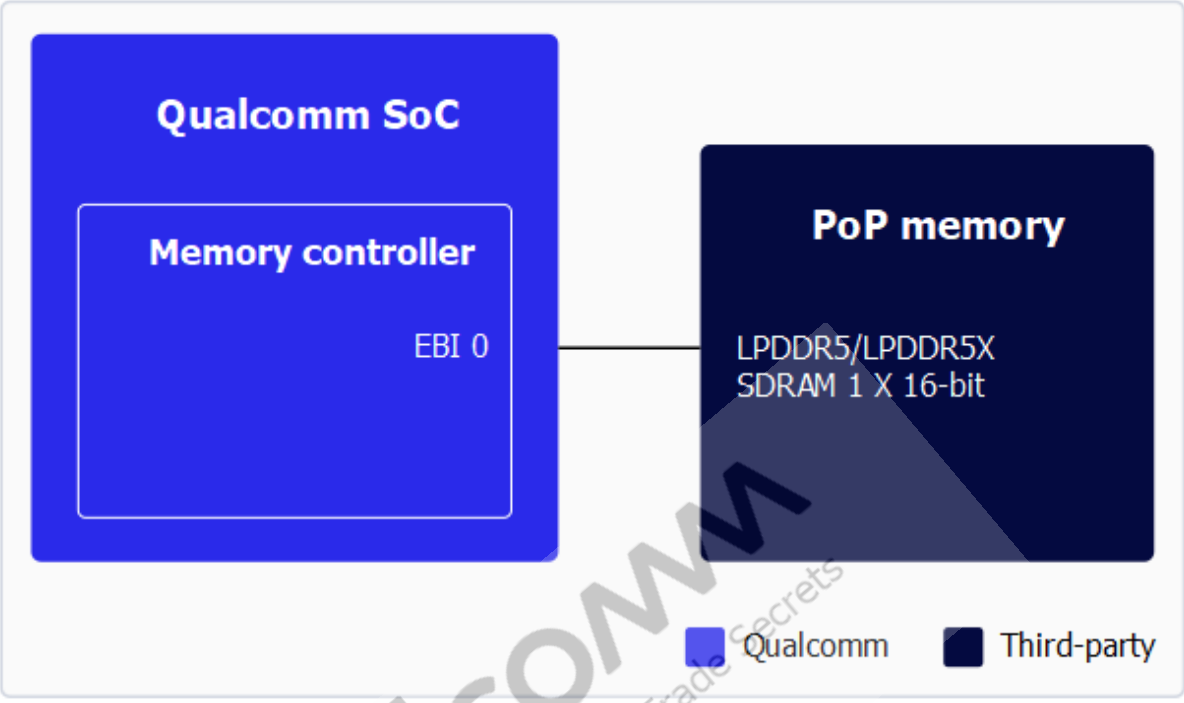


Figure : SW6100 DRAM memory channel

2.2 Enable DRAM

The DDR system firmware (DSF) is an embedded software code that allows the DRAM functionality. The eXtensible Boot Loader (XBL) image includes the DSF, which initializes the memory subsystem and runs during the boot sequence. The software works with all the DRAM components listed in the Qualcomm preferred vendor list (PVL). For information about using custom DRAM options from the PVL, see Customize and validate DRAM components.

The software doesn't require any extra modifications or configurations for the PVL DRAM components, because it automatically programs all required settings and timings during the bootup process.

2.3 Next steps

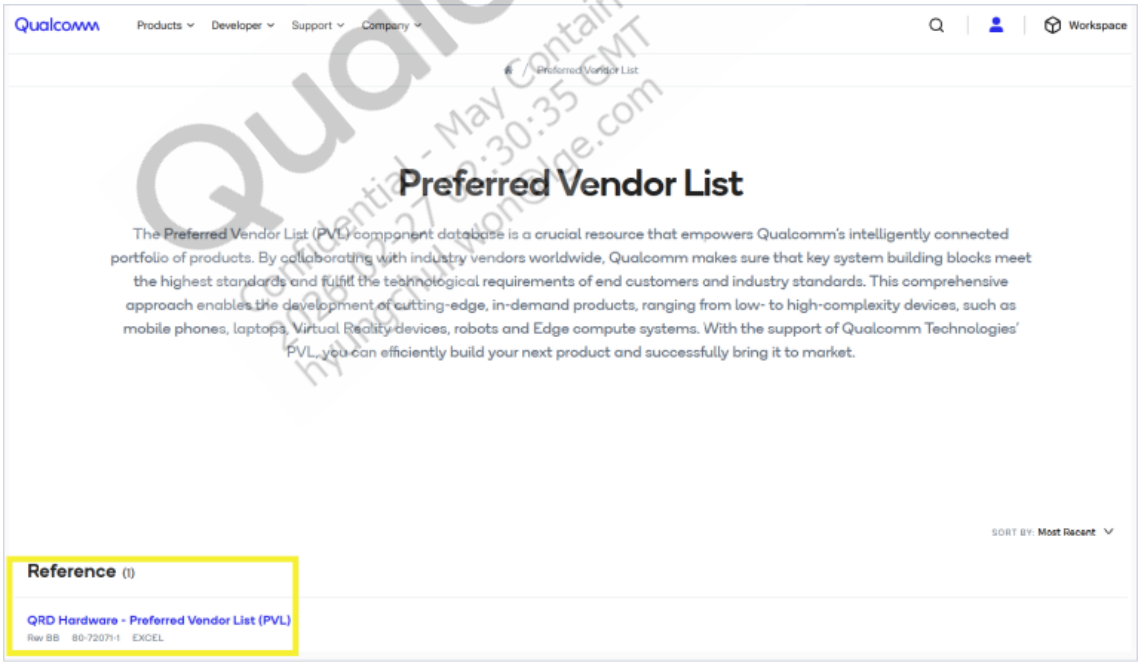
- Customize and validate DRAM components

3 Customize and validate the DRAM components

Qualcomm verifies and lists all DRAM components that work with its chip products in the preferred vendor list (PVL) component database.

To find DRAM components in the PVL, do the following:

1. Open the PVL component database.
2. Sign in to your account.
3. In the **Reference** section, select **QRD Hardware - Preferred Vendor List (PVL)** to download the Excel file.



4. Open the downloaded Excel file and go to the **HWComponents** tab.
5. Apply an autofilter to the **Type** column, select **Memory**, and then select **OK** to confirm.

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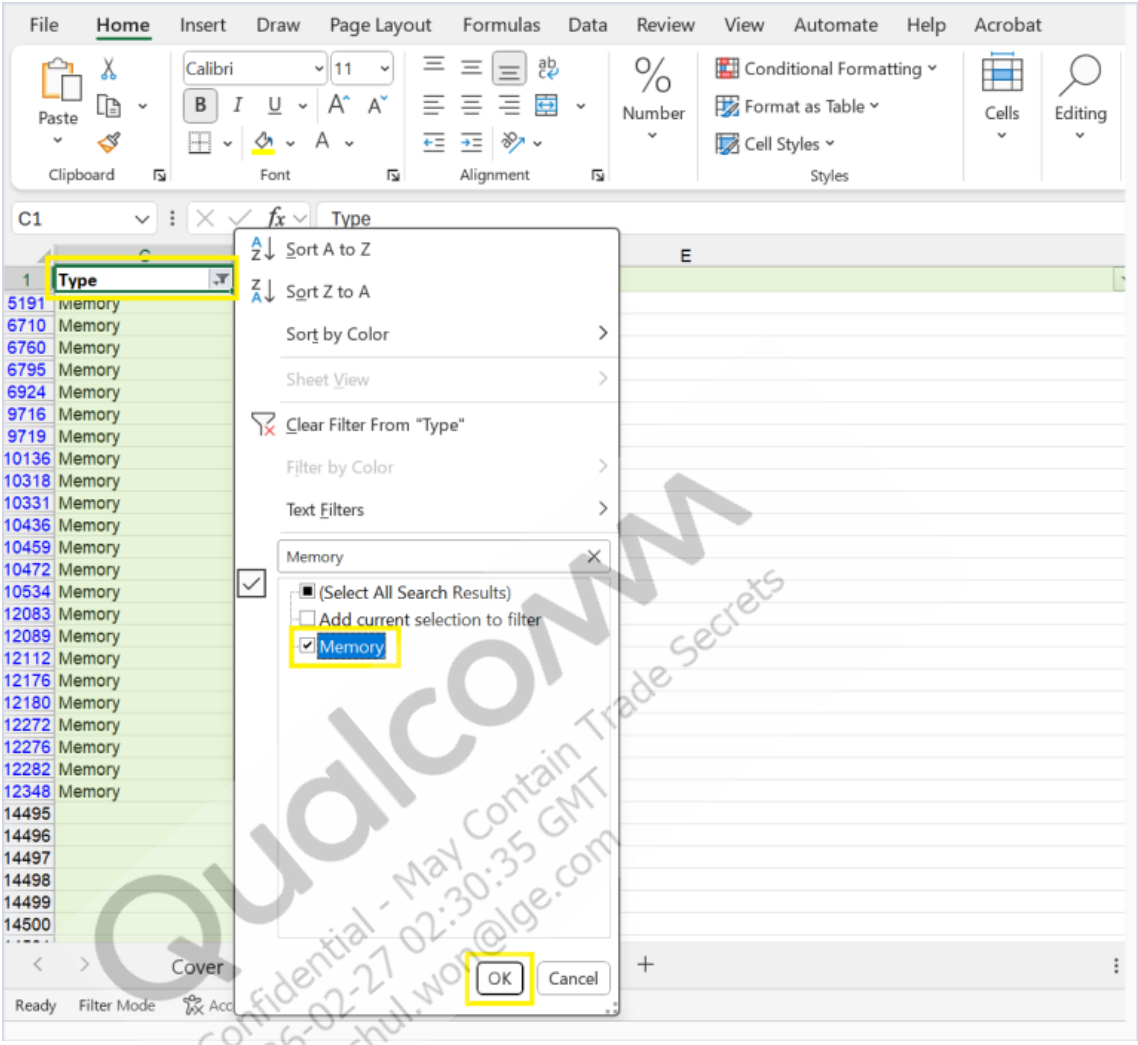

6. Choose a specific <Chipset> in the **Chipset** column and then select **OK** to apply the autofilter option. For example, replace <Chipset> with SW6100.

Note: The next release will include details about memory components supported for the SW6100 chipset PVL.

If you intend to use a different DRAM component that's not listed in the PVL for your product, contact Qualcomm for software support.

Qualcomm reviews your validation request, authenticates the component, and provides the appropriate software. After validation, use the provided software and tools such as the Qualcomm memory validation suite (QMVS). Use the QMVS tool to run a stress test and validate the DRAM.

Note: Support for SW6100 QMVS will be available soon.

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4 Troubleshoot DRAM issues

The debug information guides you to check any issues during DRAM initialization from logs and monitor the DRAM usage information during runtime.

Qualcomm provides a sample log and an example command output to help with debugging.

Note: For any DRAM-related issues, such as initialization failures, kernel panics, bit flips, corruptions, or if you are working with board designs featuring new DRAM, contact Qualcomm for software support.

4.1 DRAM initialization

The DRAM initializes sequentially during the XBL stage of bootup.

The following table lists the sequence of DRAM initialization during bootup as indicated in the serial universal asynchronous receiver/transmitter (UART) logs.

Table: DRAM initialization log entry and description

Log entry	Description
`sbl1_ddr_init, Start`	Starts DRAM initialization
`<LP5/LP5X> DDR detected`	Detects a specific DRAM component
`Manufacturer ID = <XXXX>, Device Type = <XXXX>`	Provides the manufacturing ID and device type of the DRAM
`Rank 0 size = <XXXX> MB, Rank 1 size = <XXXX> MB`	Indicates the detection of the DRAM density
`Max enabled DDR Freq = <XXXX> MHz`	Shows the maximum DRAM clock frequency
`DDR - Image Load, Start`	Completes DRAM initialization and initiates image loading on the DRAM
`DDR Frequency, <XXXX> MHz`	Sets the DRAM clock frequency to <XXXX> MHz before exiting the XBL. After the XBL exits, you can adjust the DRAM frequency to meet the requirement.

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4.2 DRAM usage

To monitor real-time DRAM usage and available memory during runtime, run the following command:

```
adb root
adb shell
vienna64:/ cat /proc/meminfo
```

The following is a sample output:

```
vienna64:/ cat /proc/meminfo
MemTotal: <XXXX> kB
MemFree: <XXXX> kB
MemAvailable: <XXXX> kB
```

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Note: To update existing DRAM settings for debugging, see QDUTT eCDT User Guide.

5 References

5.1 Related documents

Title	Number
Qualcomm Technologies, Inc.	
QDUTT eCDT User Guide	80-NR058-9
Resources	
JEDEC standard	https://www.jedec.org/category/technology-focus-area/mobile-memory-lpddr-wide-io-memory-mcp
Preferred vendor list	https://www.qualcomm.com/preferred-vendor-list

5.2 Acronyms and terms

Acronym or term	Definition
ADB	Android debug bridge
DRAM	Dynamic random-access memory
DSF	DDR system firmware
LPDDR	Low-power double data rate
PoP	Package-on-package
PVL	Preferred vendor list
QMVS	Qualcomm memory validation suite
XBL	eXtensible Boot Loader

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